

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Module	Collects rainfall, river level, and other flood-related inputs from the user.	Frontend / Input	Done	1
2	Data Preprocessing	Cleans, validates, and prepares the dataset for model training.	Preprocessing	Done	1
3	ML Model Training	Trains a machine learning model using historical flood data.	ML Engine	Done	1
4	Flood Prediction	Predicts the probability or risk of flooding based on input data.	Prediction Module	Done	1
5	Risk Classification	Classifies results into Low, Medium, or High flood risk.	Decision Logic	Done	1
6	Result Visualization	Displays prediction results and graphs for better understanding.	Visualization	Done	1
7	Weather API Integration	Fetches real-time weather or rainfall information.	API Integration	Pending	0.5
8	Alert Notification System	Sends warning alerts when flood risk is high.	Notification Module	Pending	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	5

Additional / Nice-to-Have Features	2
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input forms, result display
2	Backend / Logic	3	Preprocessing, prediction, risk classification
3	Database / Storage	1	Dataset storage
4	API / Integration	1	Weather API
5	Security / Authentication	0	Not Implemented